

1. Administrative & Study Identification

Full Study Title: Study of Pembrolizumab (MK-3475) in Participants With Advanced Solid Tumors (MK-3475-158/KEYNOTE-158)
Official Title: A Clinical Trial of Pembrolizumab (MK-3475) Evaluating Predictive Biomarkers in Subjects With Advanced Solid Tumors (KEYNOTE 158) **Short Title/Acronym:** KEYNOTE-158 **Protocol Number:** MK-3475-158 **NCT Number:** NCT02628067 **Posting ID:** 788749726026574966 **Sponsor/Company:** Merck **Investigational Product:** Pembrolizumab (MK-3475) **Phase of Development:** PHASE2 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Merck Clinical Director / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the Objective Response Rate (ORR) of pembrolizumab in participants with multiple types of advanced solid tumors, as assessed by independent central radiologic review per RECIST v1.1. **Secondary Objectives:** To evaluate the Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), and to assess the safety and tolerability profile of pembrolizumab in this patient population.

2.2 Background & Rationale To address the clinical need for effective treatments in patients with advanced (unresectable and/or metastatic) solid tumors who have progressed on standard of care therapy (or are intolerant to therapies known to provide clinical benefit). Pembrolizumab, an anti-PD-1 monoclonal antibody, has shown robust antitumor activity. This study evaluates pembrolizumab across multiple rare and common tumor cohorts, aiming to identify and validate predictive biomarkers, such as high microsatellite instability (MSI-H), mismatch repair deficiency (dMMR), and high tumor mutational burden (TMB-H), that correlate with response to immune checkpoint inhibition.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Basket Trial / Multicohort) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Nonrandomized)

3.2 Planned Enrollment & Duration Targeted Enrollment: 1,609 participants across multiple cohorts. **Number of Sites:** Global

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically-documented advanced solid tumor of one of the following types (Disease Areas: Advanced Cancer, Advanced Solid Tumors): * Anal Carcinoma / Anal Cancer (Squamous Cell) * Biliary Cancer / Adenocarcinoma (Cholangiocarcinoma, Bile Duct Cancer) * Neuroendocrine Tumor (including Carcinoid Tumor) of the lung, appendix, small intestine, colon, rectum, or pancreas * Endometrial Carcinoma / Endometrial Cancer (excluding sarcomas/mesenchymal tumors) * Cervical Carcinoma / Cervical Cancer (Squamous Cell) * Vulvar Carcinoma / Vulvar Cancer (Squamous Cell) * Small Cell Lung Carcinoma / Small Cell Lung Cancer (SCLC) * Mesothelioma * Thyroid Carcinoma / Thyroid Cancer * Salivary Gland Carcinoma / Salivary Gland Cancer / Salivary Cancer (including Parotid Gland Cancer) * Colorectal Carcinoma (CRC) that is dMMR/MSI-H (must have failed at least 2 lines of therapy; Chinese descent subset applicable) * Any advanced solid tumor (except CRC) which is MSI-H, or any advanced solid tumor (excluding dMMR/MSI-H) that is TMB-H (≥ 10 mut/Mb). 3. Progression of tumor or intolerance to at least one line of standard of care systemic therapy (not treatment naïve). 4. Provision of a tumor tissue sample for study analyses. 5. Radiologically-measurable disease per RECIST 1.1 and an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 within 3 days prior to the first dose. 6. Life expectancy of at least 3 months and adequate organ function. 7. Females of childbearing potential must use adequate contraception for 120 days post-treatment.

4.2 Exclusion Criteria 1. Prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2, or other immunomodulating agent targeting T-cell co-stimulation or checkpoint pathways. 2. Active autoimmune disease requiring systemic treatment in the past 2 years. 3. Diagnosis of immunodeficiency or receiving systemic steroid/immunosuppressive therapy within 7 days prior to the first dose. 4. Known active central nervous system (CNS) metastases, carcinomatous meningitis, or known glioblastoma multiforme of the brain stem. 5. History of noninfectious pneumonitis/interstitial lung disease requiring steroids or current pneumonitis. 6. Known history of HIV, active Hepatitis B or C, or active tuberculosis (TB). 7. Received live vaccine within 30 days of planned start of study treatment. 8. Had an allogenic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Product:

Pembrolizumab Trade Name: Keytruda **ATC Code:** L01FF02 **Dosage/Frequency:** Pembrolizumab 200 mg every 3 weeks (Q3W). **Route of Administration:** Intravenous (IV) infusion over 30 minutes. **Duration of Treatment:** Up to 35 cycles (approximately 2 years) or until documented disease progression, unacceptable toxicity, investigator decision, or patient withdrawal. **EMA Documentation:** [Keytruda EPAR Product Information](#)

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms or treatment-related toxicities. **Prohibited:** Concomitant systemic corticosteroids (except for managing immune-related adverse events), live vaccines within 30 days prior to first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy/tissue submission, Baseline imaging
Baseline	Day 1	First Dose of Pembrolizumab, Vital Signs, Safety Labs
Interim	Every 3 Weeks (Day 1 of each cycle)	Pembrolizumab administration, Safety/AE monitoring. Tumor imaging every 9 weeks for the first year, then every 12 weeks.
End of Study	Up to 35 Cycles (~2 Years) or Progression	Final safety labs, survival follow-up (up to 120 days post-treatment for contraception monitoring), End of Treatment assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), defined as the percentage of participants who have a Complete Response (CR) or a Partial Response (PR). **Measurement Method:** Assessed per modified RECIST 1.1 (maximum 10 target lesions and maximum 5 per organ) by independent central radiologic review.

7.2 Secondary & Exploratory Endpoints 1. Duration of Response (DOR) 2. Progression Free Survival (PFS) 3. Overall Survival (OS) 4. Safety and tolerability profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each clinical visit for treatment-related adverse events (TRAEs). **Serious Adverse Events (SAEs):** Reported promptly according to regulatory requirements, specifically monitoring severe hypersensitivity ($\geq$ Grade 3) to pembrolizumab and/or its excipients.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The efficacy analysis population includes all participants. Participants with unknown or missing response information will be treated as non-responders. **Sample Size Justification:** Enrollment of up to 1,609 participants ensures adequate power to assess ORR with a precise 95% Confidence Interval across specific cohorts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical site monitoring, source data verification, and independent central review for radiologic endpoints.

11. Study Identifiers & Governance

Primary ID: MK-3475-158 **Registry Number:** NCT02628067 **Posting ID:** 788749726026574966 **Sponsor/Lead Organization:** Merck **Study Title:** KEYNOTE-158

12. Clinical Framework (Oncology Specific)

Preferred UMLS Name: Advanced Solid Tumors **Primary Condition:** Advanced (unresectable/metastatic) Solid Tumors **Target Condition Deep Dive:** Carcinoid Tumor * **Name / Preferred Name:** Carcinoid Tumor * **Semantic Type:** Neoplastic Process * **Definition:** A usually small, slow-growing neoplasm composed of islands of rounded, oxyphilic, or spindle-shaped cells of medium size, with moderately small vesicular nuclei, and covered by intact mucosa with a yellow cut surface. The tumor can occur anywhere in the gastrointestinal tract (and in the lungs and other sites); approximately 90% arise in the appendix. It is now established that these tumors are of neuroendocrine origin and derive from a primitive stem cell. * **MeSH Code:** D002276 * **HPO Code:** HP:0100570 * **SNOMED ID:** 312154009 * **SNOMED Hierarchy:** 363359005 | Neoplasm of neuroendocrine tissue (disorder) | Neoplasm of neuroendocrine tissue (disorder)

13. Study Procedures & Methodology

Management Pathway: Intravenous antineoplastic therapy pathway **Education Protocol (HCP):** Identification, grading, and management of immune-mediated adverse events (irAEs). **Quality Audit**

Mechanism: Independent Central Radiologic (ICR) Review of scans to confirm RECIST 1.1 status.

14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years of active treatment (35 cycles), followed by survival follow-up. **Onsite Visit Cadence:** Every 3 weeks (Q3W) for dosing and clinical assessment. **Checklist Review Interval:** Radiologic tumor imaging every 9 weeks for the first 12 months, and every 12 weeks thereafter.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Investigator Name]	_____	03-Apr-2026	Clinical Operations Lead	[Operations Lead Name]	_____	03-Apr-2026
Lead Statistician	[Statistician Name]	_____	03-Apr-2026				